

Cycling Infrastructure Historical Analysis Pipeline

Legend

Pipeline Step

Data / File

Key Output

External Source

Loops over Cities × Years × LTS

- Run osmactive on each city PBF
- Extract cycling infrastructure (CI) edges
- Classify CI type: Separated / Painted / Mixed / Pedestrian (infra5)
- Loop: 3 years per city

- Combine trips across years per LTS
- Calculate circuitry (route/linear distance)
- Filter to OD pairs found in ALL years
- Generate comparison plots (ECDF, scatter)
- Generate overline maps (stplanr)
- Loop: 4 LTS levels per city

- Read final_city_estimations.csv
- Generate summary metric charts across LTS levels (distance, CI %, accessibility, found_routes)

- Read CI geopackages for each year
- Generate tmap output coloured by costum CI type (osmactive)

config.R
(cities, years, n_od_pairs)

city_list.txt
(city, lat, lon, population)

Geofabrik OSM.pbf
(Historical: 2016, 2021, 2026)

Global Building Atlas
(LOD1 heights)

01_city_buffers.R

- Read city coords from city_list.txt
- Create 10 km buffer polygon (EPSG:3857)
- Export bounding box (.txt)

city_10km.gpkg

city_bbox.txt

- Fetch buildings from S3 parquet (DuckDB)
- Clip to city polygon
- Sample 20k origins (uniform) & destinations (weighted by volume [m³])
- Export as .gpkg

02_od_data.R

origins.gpkg
(20k points + volume)

destinations.gpkg
(20k points + volume)

- For each city × year:
- Download historical Geofabrik .pbf (if missing)
- Crop to city bbox via osmium
- Loop: 3 years per city

03_historical_osm.R

city_YY.osm.pbf
(cropped, per year)

04_ci_osmactive.R

city_ci_osmactive_YYMMDD.gpkg
(CI edges + infra5 type)

05_r5r_routing.R

trips_city_YY_ItsN.rds
(detailed itineraries, sf)

city_YY_Its.gpkg
(LTS edge network)

routing_summary.csv
(city, year, Its, found_routes)

- Build r5r network from cropped .pbf
- Run detailed_itineraries() on 20k OD pairs
- Export LTS network edges (.gpkg)
- Log found_routes to routing_summary.csv
- Loop: 3 years × 4 LTS levels per city

06_accessibility.R

- Run r5r::accessibility() for 4 LTS levels
- Opportunities: building volume, cutoff = 15 min
- Add access_15min_vol to routing_summary.csv
- Loop: 3 years × 4 LTS levels per city

07_analysis.R

cumulative_distance_ItsN.png
circuitry_density_ItsN.png
distance_change_histogram_ItsN.png

overline_map_ItsN.png
routing_differences_ItsN.csv

routing_summary.csv (updated)
+ access_15min_vol per city/year/lts

08_final_metrics.R

- Load itineraries + LTS edge network + CI layer
- Pre-calc linear distances (OD pairs lookup)
- Compute: avg_distance, avg_circuitry, avg_dist_change_pct
- Compute: pct_ci_route, pct_ItsN (route-level)
- Compute: total_road_m, total_ci_m, pct_ci_total
- Join routing_summary (found_routes + access_15min_vol)
- Incremental upsert into final_city_estimations.csv
- Loop: 3 years × 4 LTS levels per city

final_city_estimations.csv
(per city)

09_plot_metrics.R

metric_plots/
(PNG charts per city/metric)

10_ci_maps.R

ci_map_YYMM.png
(per city/year)

11_tidy_up.R

- Remove large temporary files (cropped .pbf, mapdb caches)
- Compress .rds outputs